MetaFrontierLab Benchmark Report

Generated from: results/benchmarks_fixed_smoke

Replications per scenario: 1

Methods benchmarked: tbema, exact_baseline, dersimonian_laird, henmi_copas, metafor_trimfill_external, metafor_selmodel_external, copas_selection_external, robma_bibma_external

Executive Summary

- Best overall RMSE in this run: **dersimonian_laird** at 0.280.
- Highest observed coverage: **dersimonian_laird** at 1.000.
- Fastest method: **dersimonian_laird** at 0.001 seconds per fit.
- Methods with incomplete runs: metafor_selmodel_external, copas_selection_external, exact_baseline.
- Runtime values for external R methods include adapter startup and package-load overhead, so they reflect end-to-end benchmark cost rather than pure algorithm time.

Overall Method Ranking

npted_runs	successful_runs	invalid_ok_runs	skipped_runs	error_runs	success_rate	bias	mean_absolute_error	rmse	coverage_95	mea
4	4	0	0	0	1.000	-0.159	0.230	0.280	1.000	
4	4	0	0	0	1.000	-0.161	0.247	0.298	0.750	
4	4	0	0	0	1.000	-0.039	0.351	0.353	0.750	
4	4	0	0	0	1.000	0.055	0.331	0.355	1.000	
4	4	0	0	0	1.000	-0.565	0.579	0.673	0.750	
4	3	0	0	1	0.750	-0.064	0.199	0.251	1.000	
4	3	0	0	1	0.750	-0.060	0.427	0.436	0.667	
4	3	0	0	1	0.750	-0.490	0.495	0.604	0.667	

Scenario Highlights

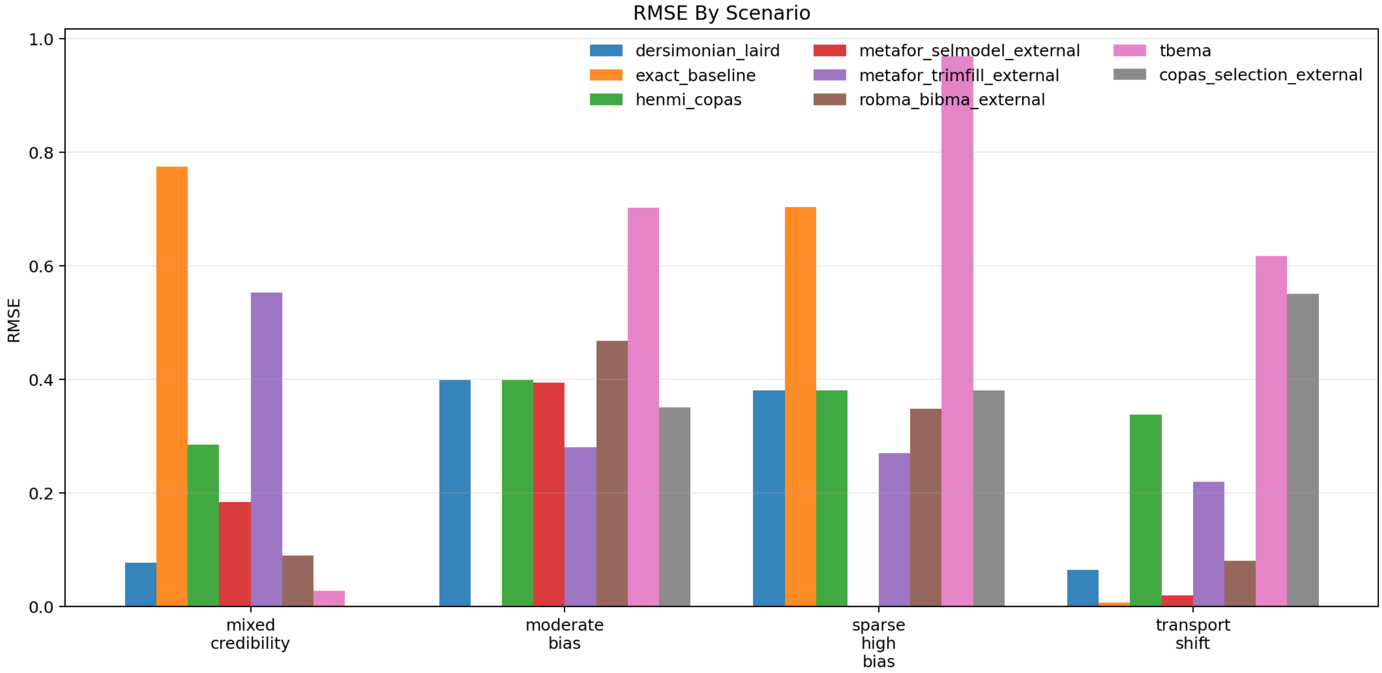
- **mixed_credibility**: best RMSE was tbema (0.028); highest coverage was tbema (1.000). Incomplete runs: copas_selection_external.
- **moderate_bias**: best RMSE was metafor_trimfill_external (0.281); highest coverage was metafor_trimfill_external (1.000). Incomplete runs: exact_baseline.
- **sparse_high_bias**: best RMSE was metafor_trimfill_external (0.270); highest coverage was metafor_trimfill_external (1.000). Incomplete runs: metafor_selmodel_external.
- **transport_shift**: best RMSE was exact_baseline (0.006); highest coverage was exact_baseline (1.000).

Scenario Table

	replications_total	successful_runs	invalid_ok_runs	skipped_runs	error_runs	success_rate	bias	rmse	coverage_95	mean
ction_external	1	0	0	0	1	0.000				
simonian_laird	1	1	0	0	0	1.000	0.077	0.077	1.000	
xact_baseline	1	1	0	0	0	1.000	-0.774	0.774	1.000	
henmi_copas	1	1	0	0	0	1.000	0.285	0.285	1.000	
odel_external	1	1	0	0	0	1.000	0.184	0.184	1.000	
rimfill_external	1	1	0	0	0	1.000	0.553	0.553	1.000	
ibma_external	1	1	0	0	0	1.000	0.090	0.090	1.000	
tbema	1	1	0	0	0	1.000	0.028	0.028	1.000	
ction_external	1	1	0	0	0	1.000	-0.350	0.350	1.000	
simonian_laird	1	1	0	0	0	1.000	-0.399	0.399	1.000	
xact_baseline	1	0	0	0	1	0.000				
henmi_copas	1	1	0	0	0	1.000	-0.399	0.399	0.000	
odel_external	1	1	0	0	0	1.000	-0.394	0.394	1.000	
rimfill_external	1	1	0	0	0	1.000	-0.281	0.281	1.000	
ibma_external	1	1	0	0	0	1.000	-0.468	0.468	0.000	
tbema	1	1	0	0	0	1.000	-0.702	0.702	0.000	
ction_external	1	1	0	0	0	1.000	-0.380	0.380	1.000	
simonian_laird	1	1	0	0	0	1.000	-0.380	0.380	1.000	
xact_baseline	1	1	0	0	0	1.000	-0.704	0.704	0.000	
henmi_copas	1	1	0	0	0	1.000	-0.380	0.380	1.000	
odel_external	1	0	0	0	1	0.000				
rimfill_external	1	1	0	0	0	1.000	-0.270	0.270	1.000	
ibma_external	1	1	0	0	0	1.000	-0.348	0.348	1.000	
tbema	1	1	0	0	0	1.000	-0.968	0.968	1.000	
ction_external	1	1	0	0	0	1.000	0.550	0.550	0.000	
simonian_laird	1	1	0	0	0	1.000	0.064	0.064	1.000	
xact_baseline	1	1	0	0	0	1.000	0.006	0.006	1.000	
henmi_copas	1	1	0	0	0	1.000	0.338	0.338	1.000	
odel_external	1	1	0	0	0	1.000	0.019	0.019	1.000	
rimfill_external	1	1	0	0	0	1.000	0.220	0.220	1.000	
ibma_external	1	1	0	0	0	1.000	0.081	0.081	1.000	
tbema	1	1	0	0	0	1.000	-0.616	0.616	1.000	

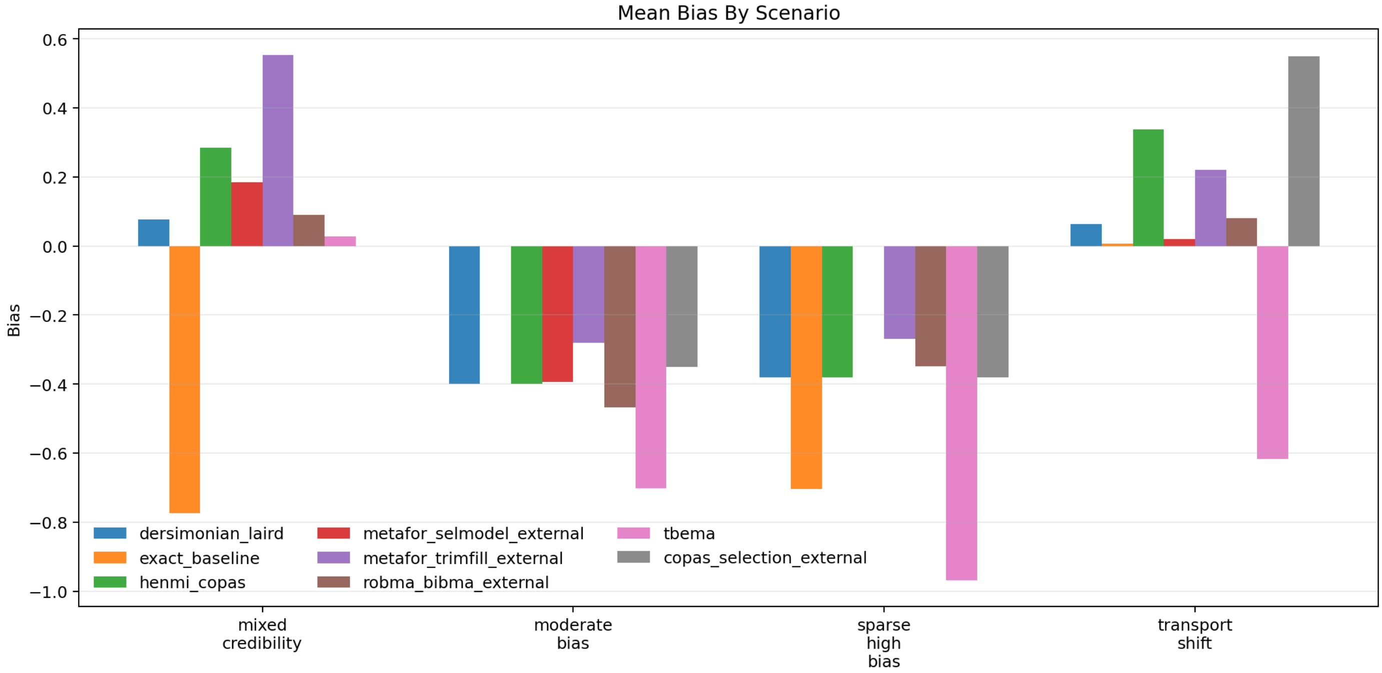
RMSE by Scenario

Figure generated from the benchmark summary.



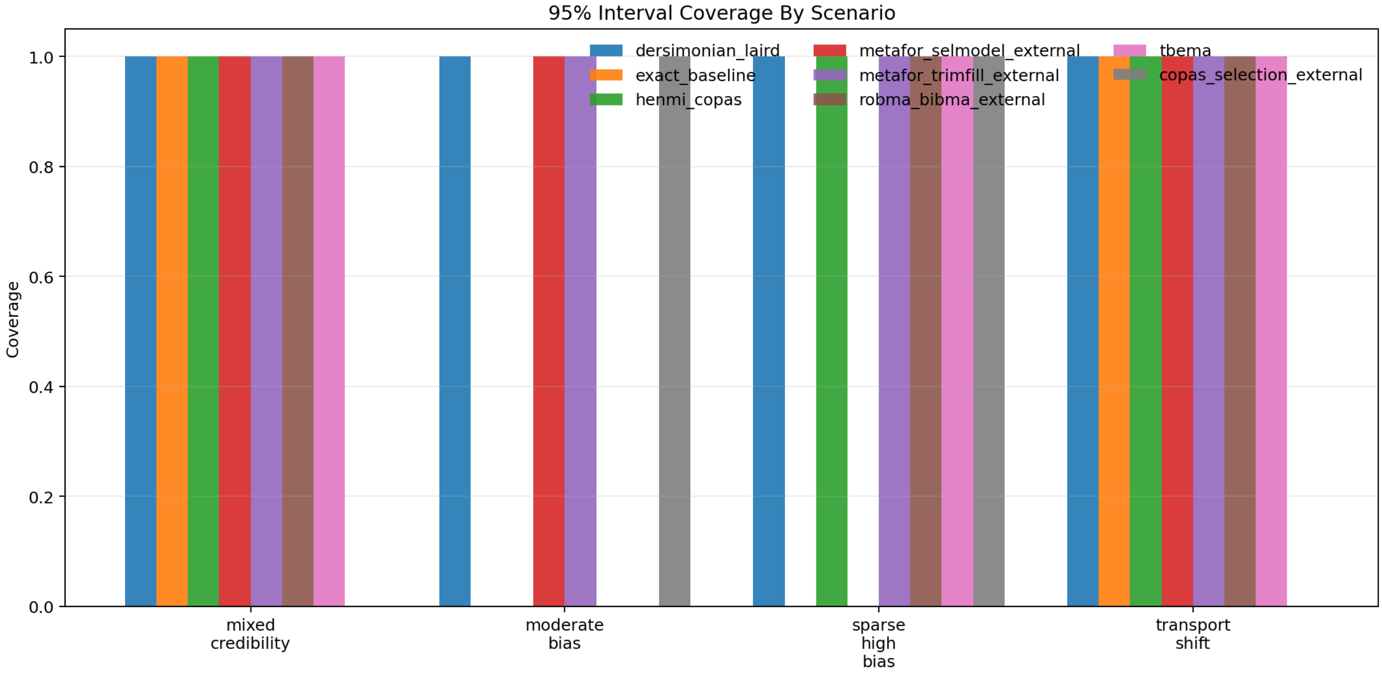
Bias by Scenario

Figure generated from the benchmark summary.



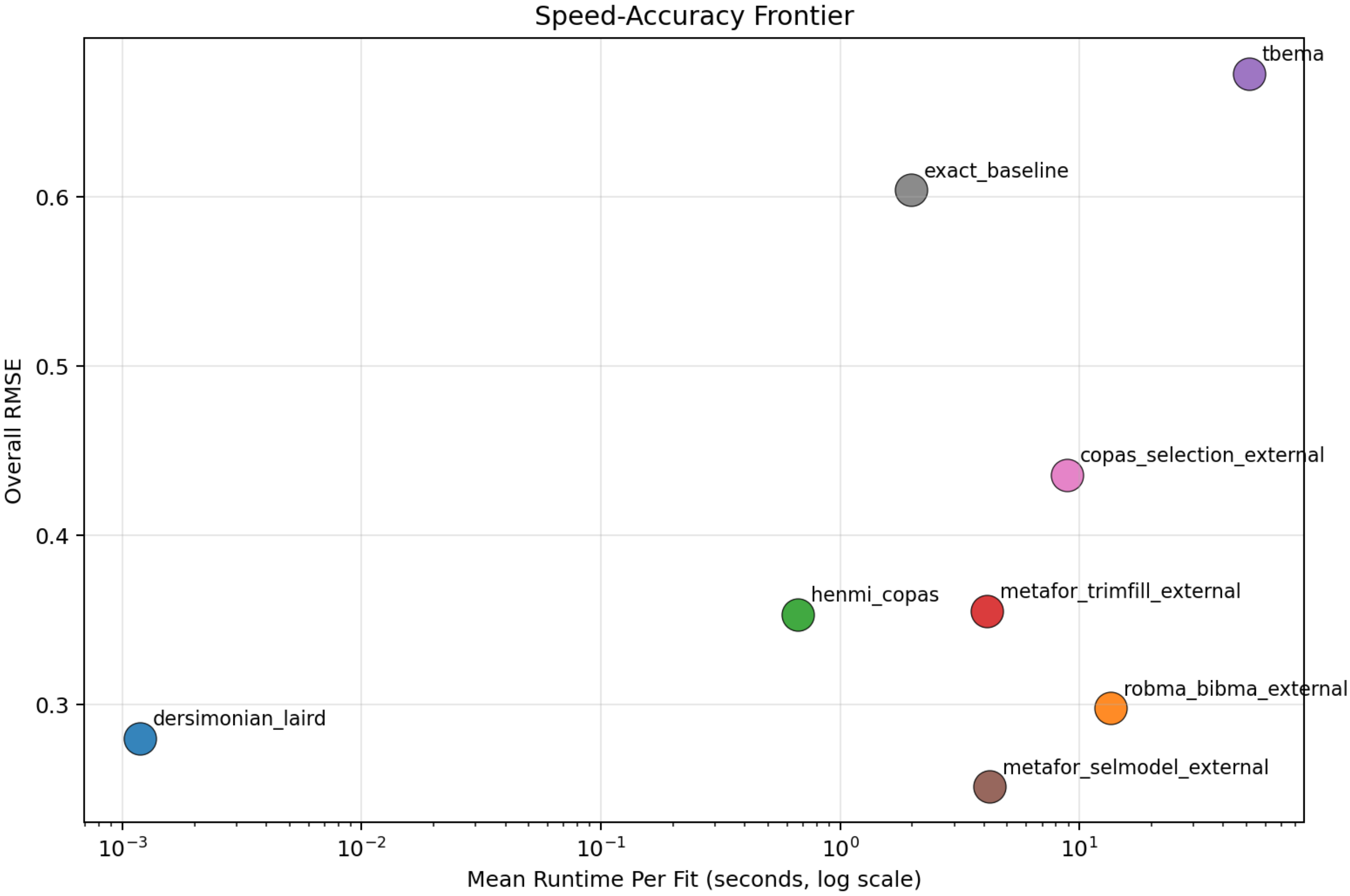
Coverage by Scenario

Figure generated from the benchmark summary.



Speed-Accuracy Frontier

Figure generated from the benchmark summary.



Methods Appendix

- TBEMA: exact sparse-data meta-analysis with stacked bias tempering, transport weighting, and ridge-regularized target-aware meta-regression.
- DerSimonian-Laird: standard normal-normal random-effects benchmark.
- Henmi-Copas: publication-bias-robust interval method.
- metafor trimfill: trim-and-fill adjustment from metafor.
- metafor selmodel: step-function selection model from metafor.
- Copas selection: Copas model from metasens.
- RoBMA BiBMA: Bayesian model averaging for binary outcomes, using the fast spike-and-slab algorithm.